

ActInf Textbook Group ~ Cohort 3 ~ Meeting 11 (Chapter 5, part 2)

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Hello, it is April 5th, 2023, and we're in our second discussion of Chapter 5 in Cohort 3.

So this is going to be our final recorded session for the first half of the textbook, Chapters 1-5.

And then next week, we'll have an unrecorded session.

We will have time to fill out the feedback form for those who haven't yet.

We'll discuss project ideas, which people can also add ideas to.

We'll just talk a little bit more broadly and informally about the textbook and about the textbook group.

And then we will have an inter-semester break, or it's not a semester, but an intermission.

And we'll pick up when Cohort 4 begins, we'll pick up with the second half of Cohort 3 going through, just as previous cohorts did, Chapter 6, 7, 8, 9, 10, and so on.

So, we're in our second discussion of Chapter 5.

I'll head over there now, and we'll begin just with anybody in the chat, or via raising their hand, or just unmuting.

How does Chapter 5 sit with you?

What are any thoughts or questions about Chapter 5 that you want to just raise at the outset?

Chapter 5

Jonathan?

Yeah, I sort of feel that while, in some sense, it says that you don't need to have digested Chapter 4 to understand Chapter 5,

it still feels like there's a lot of background there, which is really important.

And the lack of technical clarity from Chapter 4, I sort of feel there's a bit of a hangover from that into Chapter 5.

The technical debt hangover.

Yep.

Yes.

Okay.

Thank you, Maria.

I was a bit frustrated with this chapter because I thought it would be more insightful, but it seemed to me it was just more of the same.

Bunch of equations narrated and plotted.

Yes.

I guess the apologist's view would be Chapter 4 was equations and plots in general.

And Chapter 5, in contrast, is applying it to the specific mammalian nervous system examples and to empirical studies.

However, broadly, I totally agree with that.

Any other thoughts or primary reflections on Chapter 5?

Any aspect of it or some part of the discussion or even any other active inference learnings over the last

week or two that come to your attention when you're in this setting?

Okay.

So, again, raise your hand or write in the chat, Terry, you wrote, I was struck by the similarity to neural networks.

You could describe a little more if you want to unmute, or we can see how that might come into play.

So, in the previous chapter 10 discussion from cohort three, we were playing with the idea of reading chapter 10 first to provide a lot of the conceptual scope and the historical connections and similarities and dissimilarities of active inference with other topics.

And so, it's interesting to note, neural network is described in the citations and twice in chapter 10 in this paragraph.

But there's a lot of context that is kind of held out till chapter 10.

But the intuitions that you're having along the way are very good.

Any other just general thought or a question on five?

Otherwise, we will look to the questions and address those which we haven't addressed previously and then continue on through.

Okay.

So, continue on this neural network just while we're here.

Let's use the figure 5.5.

So, here we have 5.5.

Oh.

The cortical system.

The limbic or the midbrain system associated with policy selection.

And the spinal motor execution arc using this differential between perceptions and set points.

And so, then, Terry, you wrote,

the series of nodes passing information along a chain with inhibitory mechanisms alongside.

Yet, neural networks use transformers while active inference uses free energy.

Well, great topic.

To go deeper into the relationship between neural networks and free energy principle and active inference models.

In livestream 51.

I'm just going to the slides now.

The core claim is that here on the right,
we see the Markov blanket and the internal and the external states.

Sense coming in.

Perception.

Action selection resulting in embodied action.

And an effect upon the generative process, the niche, which then provides observations.

So, this is how we've been looking at it in the textbook.

On the left side, we see in a three-layer neural network,

an input layer,

one or more compute or intermediate layers,

which can have a wide range of connectivities,

and then some output layer.

So, this could be the image comes in and the classifier label comes out.

Or the classifier label comes in and the image generated comes out.

And then it acts in an environment,

which then provides another observation.

And in this paper,

they demonstrate that the variational free energy on this partition

is equivalent to the loss function on this neural network.

So, the training of this neural network

is monotonically related to the variational free energy loss on this model learning.

The loss function is the function that's being minimized to update this neural network.

And so, it's really quite a stunning conclusion

because the topologies,

so how the pieces are connected,

of these two models,

is entirely different.

So, in Livestream 51,

there's more detail on that.

Neural networks use transformers

while active inference uses free energy.

And Jonathan asks,

transformers in the post-2017 technical sense?

So, in the sort of attention is all you need,

pre-trained transformer timeline that we're on?

That's a good question.

And that feels specifically like a reinforcement learning framework,
or is it more general?

So, I meant there in terms of the diagram you were giving,
where there was a neural network
and a sort of feedback on an environment.

Yeah, this one here,

which, yeah,

it looks like a sort of RL framework,

but are you saying this is sort of more general than that?

It is more general,

as I understood the paper,

describing that for the observation sequence,

one to T ,

and the parameterization,

all the weights of the neural network.

And then analogously,

the variational free energy

of those very same observations

and the parameterization of the Bayes' graph,

these two are monotonically related

within certain classes,

quite broad classes.

Yes, it was,

I mean, it's,

I believe, a very major recent finding

because it suggests that we can have this,

like this,

on one hand,

it looks like a circuit board

and it's a lot to look at.

However,

there is a semantic interpretability.

Here we see prior,

hidden state,

B,

that's how things change through time,

hidden state,
changing through time,
hidden state,
so there's hidden states changing through time.

Here's the A matrix,
emitting observations,
or you could recognize observations,
decision making,
preferences guiding decision making.

So it's like there is an interpretability
to this Bayes' graph.

And so the idea that a neural network,
which can have an arbitrary topology
with absolutely no semantic interpretability,
kind of like any given neuron
in a real neural system,

I mean,
Jennifer Aniston neuron aside,
no neuron plays an explicit semantic role,
let alone a one-to-one semantic role.

So the idea that this kind of an aggregate system
would have a representation
where the variables are semantic
is quite a claim.

And then the notion that the training function
for both of them are monotonically related
is really important.

So not to go too far afield from the textbook,
but just neural networks were brought in.

And that's an example of some very recent research
that makes that connection go beyond intuition.

Okay.

Here's a question
that I don't think that we addressed previously,
but if anyone else has any other question,
please raise your hand or write in the chat.

So,

table 5.1

gives remarkable association
between neurotransmitters and precisions.
Should we be suspicious
that this seems so clean
or is this evidence
that the framework is a good match
for real brain functioning?
What does anyone think?
or what was their prior suspicion
before seeing the table?
And then what was their posterior suspicion
after they saw the table?
Jonathan?
And then anyone else?
Yeah.
So this was my question.
I think I was just surprised
that there was sort of this
apparent one-to-one correspondence
or at least near one-to-one correspondence.
So I think I would have expected
different neurotransmitters
to have
a more,
I guess,
just there not to be
this very clean relationship.
That was my prior.
Certainly,
these molecules
do
things
that are not only listed here.
Like serotonin,
the word means blood pressure.
Serotonin was discovered in the blood,
as was dopamine.
And it has

a variety of effects
on immune cells.

And then later
was understood
to play a role
in the central nervous system.

So
I don't think
that these
evidence provided
are suggesting
that the exclusive function
of these neurotransmitter
type molecules
is
that they're
exclusively
and exhaustively
described
by these evidences.

However,
these do provide
empirical
patterns
of
everything
from
histology
to
association
with cognitive
functions,
neuroimaging studies,
gene expression
analysis,
pharmacological
manipulation,
like

the whole
molecular
neuroscience
toolkit.

These are
quite
disparate
phenomena.

And
they're
disparate
phenomena
that on one
hand have been
linked mechanistically
to a given
molecule
and associated
receptors
and all of this.

But on the other
hand,
these phenomena
are conceptually
linked
to certain,
to the semantics
of certain
parameters
in these
statistical
models.

maps are
not
territories.

It's not
saying that
it's

exhaustively
described.
However,
if we
use this
association
as a
heuristic,
if not a
hypothesis,
and we
find that
we have
unique
explanatory
and predictive
powers,
then,
by the
standards
of
science,
it was
a valid
modeling
approach.
These are
explanations
and predictions
that would
not be
achievable
without
mapping
a molecule
to the
semantics
of a

parameter.

like,

just

searching

acetylcholine

on a

literature

search

engine,

you'll

find a

whole

variety

of

different

studies,

different

study

types,

different

outcomes,

different

species,

and so

on.

So,

that's

the

scruffy.

The

scruffy is

what the

search engine

just gives

you.

Each

paper is

unique,

every
experiment
is
unique.
And then
the
neater
than
neat
would be
like,
all
neurotransmitters
do one
thing.
And so
somewhere
that is
useful
enough
to help
us make
explanations
and predictions,
but not
so terminally
differentiated
that we
can't link
together
novel
strands
of evidence
into
explanations
and
predictions,
somewhere

bridging
or
respecting
the
neat
and
scruffy
articulation
is
active
inference
and
the
ways
that
we
can
use
statistical
models
to
connect
to
these
broad
empirical
results
on
different
chemicals.
So
should we
be
suspicious
that it's
so clean?
Well,
yes,

they're
presenting
their
arguments
like a
prosecutor.

They're
not necessarily
going into
every
caveat
or
bringing
up
studies
that
might
disconfirm
that.

So for
sure it's
connecting
dots and
not all
the dots
are being
shown and
not all
the connections
are being
shown.

but
it
moves
the
discussion
forward
about

how
aspects
of
maps
which
are to
say
cognitive
models
are
related
to
aspects
of
territory
which
include
the
actual
materiality
of the
neural
system.

So
Michael
and then
anyone
else.

Well,
Daniel,
sorry I've
been
out of
pocket
for a
week
or so
but

I'm
trying to
catch up
and the
way I'm
trying to
make sense
of what
I'm
now
landing
into
I wanted
to
kind of
validate
does this
make sense
or not
make sense
if
one were
to think
and the
way I've
been
trying to
the mental
models
that I've
been trying
to map
this
conversation
to
are
about
social

conversations
and
you could
say
one level
of activity
up from
the biological
biochemistry
and I was
thinking
if these
were
verbs
sorry
if each
of these
neurotransmitters
were nouns
verbs
adjectives
and that
sentences
were being
created
they could
be created
many different
ways
but
they're
units
of
composition
if you
will
of
message

signaling
that
are
creating
larger
sentences
at that
level
of
the
system
would
that
metaphor
make sense
or
would
it
be
missing
how
might
it
be
a
mismatch
for
what
is
happening
great
question
does
anyone
have a
thought
on
that

potentially
referencing
figure
5.5
or
any
other
area
but
any
thought
on
this
question
I'll
give a
first
thought
then
please
anyone
can
build
on
it
so
what
is
speech
really
it's
this
continuous
flow
of
like
phonemes
and

so
it
is
not
subject
object
verb
in
how
it
actually
is
like
the
materiality
of
speech
is
vibration
in
the
air
and
all
of
that
and
the
embodiment
and
so
that's
like
the
territory
of
speech
but

let's
just
say
that
we're
putting
on
our
linguist
hat
so
now
we're
going
to
think
about
certain
parts
of
speech
as
like
subject
object
verb
or noun
and adverb
and so
on
even
though
the
speaker
may
have
not
thought

about
it
that
way
so
we're
making
maps
of
sentences
as
linguists
even
if
the
speaker
didn't
use
that
map
when
they
generated
the
speech
!
does
this
whole
figure
represent?
And
beliefs
about
this
and
beliefs
about

that
lead
to
some
differential
contrasts
in light
of incoming
evidence
and
the
result
of
that
is
now
we're
going to
leave
that
little
clause
and
the
result
of
that
contrast
in light
of
incoming
evidence
is
a
decision
making
process
about

what
to
do
given
precision
on
the
decision
making
process
and
one
of
the
outcomes
of
that
decision
making
process
is
a
descending
prediction
that
contrasts
with
incoming
sensory
data
and
results
in an
action
hopefully
it's not
too
mixed

of a
metaphor
but
absolutely
these
different
compositions
connections
just like
infinite
grammar
you can
keep
on
adding
yes
and
also
in the
social
space
you can
keep
on
adding
sentences
to each
other
and make
longer
paragraphs
you can
keep
on making
longer
sentences
or lists
similarly

you can
keep
on
making
larger
generative
models
or
different
generative
models
and so
different
connections
amongst
these
units
are going
to be
described
kind of
like
by the
genre
like
that
was a
haiku
that
was a
sentence
with
no
noun
that
was a
question
that

was a
directive
so
loosely
it's a
great
working
model
to
think
about
this
as
a
graphically
arranged
sentence
that's
expressing
something
where
some
of the
nodes
some
of the
tokens
are
associated
with
more
belief
like
states
some
of the
nodes
are

associated
with
observations
some
of the
nodes
are
associated
with
capacities
for
action
and so
on
and so
we can
kind of
like
play
with
legos
play
with
language
and
compose
like
well
what if
there
were
these
three
sensory
modalities
and
two
action

possibilities
and
decision
making
was
done
this
way
that's
a
generative
model
cool
yeah
and
then
however
that
sentence
is
diagrammed
it
simply
wouldn't
be
the
actual
speech
itself
it
wouldn't
be
the
vibration
of
the
voice
box

or the
air
so
that's
like
map
territory
but
once
we're
in
the
map
then
sentences
might
have
this
kind
of
structure
or
this
could
be
the
dependency
structure
of
an
organization
or
the
dependency
structure
of
claims
in a

scientific
corpus
that's
kind
of
what
Bayes
graphs
do
they
describe
different
variables
and
their
associations
and
that
also
speaks
to
their
flexibility
like
you
could
know
all
of
these
and
solve
for
that
one
or
you
could

just
have
these
ones
and
try
to
reduce
your
uncertainty
about
those
so
that
would
be
like
a
sentence
where
you
did
hear
the
first
part
but
not
the
second
part
or
vice
versa
thank
you
dan
again

every
time
i
during
these
conversations
my mind
gets
mental
models
get
blown
open
and
thank
you
again
good
well
hopefully
they're
not
reduced
to
rabul
because
your
intuition
was
right
about
taking a
linguistic
approach
i mean
who's
to say
it's

right
it
certainly
is
compatible
and
this
is
like
a
composition
this
is a
setting
where
we
can
compose
generative
models
and
so
that's
why
sometimes
it's
like
a
little
challenging
what's
the
active
inference
answer
on
attention
what's

the
answer
on
memory
it's
like
those
are
complex
cognitive
phenomena
that play
out
differently
in
different
systems
so
if
the
framework
were
rigid
enough
to
give
an
answer
to
that
in
general
it
wouldn't
have
the
latitude
that

we
need
to
actually
describe
that
phenomena
and so
we
want
to
have
the
latitude
to
describe
different
phenomena
including
unnamed
phenomena
in
systems
we
haven't
seen
yet
which
means
that
the
framework
itself
can't
be
so
specific
about

how
certain
things
play
out
it's
mind
blowing
it's
absolutely
mind
blowing
and it's
such a
privilege
I know
I said
every
time
it's
such a
privilege
to be
in this
conversation
and
witnessing
you
make sense
of it
through
this
complex
collective
lens
when I've
been
thinking

in blocks
and
these
blocks
are
being
you know
it's
like
no
these
aren't
bricks
they're
flows
and
it
works
cause
and
effect
works
differently
and
it's
just
as I
say
mind
blowing
and
daunting
but
thank
you
thank
you
it's

definitely
team
efforts
and
we've
been
through
it
many
times
and
many
people's
insights
have
been
folded
together
does
anyone
else
in
chat
or
raising
their
hand
want
to
have
any
reflection
on
five
or
a
question
or

we
can
look
over
some
sections
and
then
in
the
last
minutes
we
can
look
ahead
to
the
second
part
of
the
book
or
whatever
people
prefer
is
there
a
neural
system
or
some
embodied
behaviors
that
people

want
to
investigate
I'll
add
in
this
question
okay
thank
you
Yana
are
there
alternative
visualizations
that better
show
the
interdependencies
that's
really
interesting
question
in
a
base
graph
the
edges
reflect
yeah
and I'll
add this
thank you
in the
base
graph

the
edges
reflect
a
certain
kind
of
statistical
connection
amongst
variables
but
this
entire
structure
it's
kind
of
like
a
web
or
like a
tensegrity
sculpture
or
like a
guitar
string
in that
the
relate
an edge
like let's
just say
it from
this
S3

to
Pi
3
but
yes
but
then
if
policy
selection
ends
up
doing
something
in
the
real
world
then
that
is
going
to
influence
future
proprioceptive
inputs
so
even
edges
that
aren't
shown
it's
not
that
they're
not

associated
with
each
other
through
the
reality
of
the
system
but
the
edges
are
showing
the
kind
of
direct
connections
again
not
the
anatomical
direct
connections
then
it would
just be
an
anatomy
textbook
but
it's
showing
the
direct
informational

or
causal
connections
so
how can
we
visualize
the
interdependencies
there's
so
many
ways
we
might
explore
but
I
really
like
that
as
something
for
us
to
think
more
about
like
one
more
traditional
way
in
terms
of
scientific

visualization

would

be

we

could

plot

S3

as

on

the

x

axis

and

you

know

mu

sub

x1

on

the

y

axis

and

then

we

could

ask

whether

there

was

any

correlation

or

mutual

information

between

those

two

variables
so
the
fact
that
they're
not
connected
it
doesn't
mean
it's
going
to
be
just
a
blur
in
fact
they
might
be
very
strongly
associated
with
each
other
but
through
some
intermediate
nodes
so
that
would
be

one
way
of
looking
at
whether
any
given
nodes
two
or
three
or
however
many
whether
they
have
associations
or
interdependencies
and
then
I guess
another
little
layer
is
the
space
of
the
counterfactual
like
some
interdependencies
might not
be revealed

unless
something
changes
in the
system
so
that
is
also
a
layer
where
two
things
might have
no
correlation
and then
you
remove
an edge
and then
all of a
sudden
two things
that
weren't
in
association
now
are
okay
I
don't
think
that's
a
complete

answer
because
there's
a lot
of work
with
visualization
to be
done
but
that
is
just
kind
of
some
aspect
of
it
and
then
just
continuing
with the
honest
question
and then
to
Jonathan
so
it
is
like
a
magnetic
coil
the
electric

field
is
not
going
along
the
coil
it
is
a
field
so
lot
to
think
about
don't
have
a
specific
thought
if
anybody
wants
to
add
something
but
yes
I
like
what
if
these
are
wires
and
then

there's
like
fields
of
influence
that
aren't
just
the
wires
but
the
connections
are
there
and
they're
relevant
but
then
there
might
be
other
dependencies
that
aren't
simply
mapped
out
on
the
wire
dependencies
um
okay
Jonathan
I'm

going to
copy
your
question
in
has
this
been
looked
at
in
hypergraph
language
rather
than
graphs
in
active
block
friends
we
looked
briefly
at
hypergraphs
we
haven't
seen
any
active
hypergraphs
but
it's
very
interesting
a
graph
through

time
so
for
example
what
if
this
right
now
there's
a
static
topology
so
even
if
there's
dynamic
information
flowing
through
this
graph
you
could
say
that
it's
a
static
graph
the
referent
is
dynamic
it's
changing
through

time
but
the
structure
is
not
changing
through
time
but
what
if
on
even
and
odd
time
steps
this
connection
existed
or not
then
that
kind
of
a
system
as far
as I
understand
is
a
hypergraph
is there
some other
aspect
to

hypergraph

jonathan

or

what

would

enabling

this

inactive

inference

help

us

do

so

hypergraph

sort

of

allow

for

more

general

types

of

edges

it

sort

of

allows

for

edges

which

specifically

go

through

vertices

kind

of

the

easiest

example
is
a
plane
routing
system
airplanes
where
if
a
single
airplane
goes
from
airport
A
to
B
to
C
it's
very
clear
that
it's
the
same
airplane
that's
gone
between
those
and
just
by
drawing
an
edge

between

A

and

B

and

B

and

C

that

doesn't

give

the

same

information

that

a

single

piece

of

information

or a

single

airplane

has

taken

a

particular

route

so it

just

allows

for

more

generalizations

of

edges

than

in

a
regular
graph
I
just
wonder
essentially
about
the
way
you
can
encode
the
types
of
information
that
move
through
a
regular
graph
and a
hypergraph
and one
might be
able to
encode
more
specific
information
in a
hypergraph
yeah
very
interesting
I'm

pretty
sure
that
it
hasn't
been
written
on
in
act
inf
a
lot
of
the
papers
seem
very
recent
like
the
one
you
point
to
is
2018
and
the
others
are
recent
so
it
would
be
an
interesting

area
one
note
on
this
to
kind
of
speak
to
the
other
side
where
the
kind
of
map
is
very
effective
with
respect
to
the
territory
for
better
and
for
worse
what
if
these
were
computers
and
these

were
the
literal
wires
connecting
them
in
that
case
you
could
design
in
silico
or
cyber
physical
systems
where
the
edges
of
the
base
graph
very
well
map
to
the
causal
structure
of
the
system
physically
so
the

kind
of
one
of
the
hardest
systems
and
one
of
the
richest
systems
is
like
trying
to
make
simple
statistical
models
of
the
brain
but
when
making
just
in
silico
systems
so
just
simple
pedagogical
models
or
intuition

pumps
or
in
certain
systems
for
digital
and
communications
systems
the
edges
can
have
a
much
more
clear
interpretation
because
as laid
out in
the
beginning
of
five
this
is
kind
of
this
is
the
tension
and
the
dream
would

be
what
if
the
actual
anatomical
connections
amongst
these
cell
types
what
if
those
were
actually
reflective
of
the
structure
of
a
base
graph
and
you'll
note
that
they're
not
exactly
the
same
like
just
here's
you know
two

to
three
and
here
we
see
that's
two
to
three
but
also
it
goes
from
here
from
from
four
to
two
but
we
don't
see
a
four
to
two
in
the
tissue
so
but
it's
close
enough
but

clearly
not
close
enough
to
merit
calling
this
simply
layer
five
and
I guess
to
the
electric
field
these
connections
are
the
neural
connections
what
if
there
are
glial
connections
what
if
there
are
electric
field
influences
like
what

the
EEG
measures
are
some
of
the
electrical
phenomena
of
these
cells
and
actually
just a
specific
kind
of
electrical
phenomena
of
these
cells
the
EEG
outside
of
the
head
but
those
electrical
signals
are not
only
outputs
like
epiphenomenal

to neural
activity
they're also
inducing
different
changes
in neural
activity
as well
so that
there's also
physical
edges
that aren't
shown
just
you know
complex
systems
are
complex
and
they
engage
in
multiple
mechanistic
ways
which is
why it's
important to
have maps
of them
Ali
yeah
well
actually going
back to

Yana's
question
and also
to
touch on
what
Jonathan
just
mentioned
well
in fact
when we're
talking about
interaction
obviously
there are
many
different
conceptions
of
this
term
interaction
so
we can
talk about
physical
interaction
which is
basically
I mean
for the
most part
is
undertaken
the
fields
so

we can
also
term it
as a
kind of
field
interaction
but
there are
some
other
conceptions
of
interaction
actually
there are
many
others
but
probably
one of
the most
important
conceptions
for
our
purposes
is
information
theoretic
interaction
so
when we're
talking
about
message
passing
or

this
conception
of
interaction
through
I mean
information
passing
or information
theoretic
conception
of it
we're not
necessarily
deciding
on
what
exactly
is
the
medium
of
interaction
or
transmission
because
it can
be
abstracted
away
this
medium
of
transmission
or
interaction
and we
can

just
look at
the
mathematical
framework
of
or the
mathematical
structure
of those
information
so for
instance
say
IIT
framework
or
integrated
information
framework
in which
well
in
information
integrated
information
theory
it's unable
to distinguish
a highly
interdependent
set of
components
dominated
by
redundant
relations
from an

equally
strongly
interacting
set of
components
exhibiting
a kind
of strong
synergy
between
them
so
this is
actually
kind of
a limitation
for
IIT
for some
people
because
since
redundancy
indicates
that
although
strong
interaction
is needed
to ensure
that a
component
is a
kind of
repetition
of the
other
components

it does
not imply
that
the
set
of
components
support
collective
emergent
behavior
which is
entirely
different
from that
of the
individual
components
so
in
recent
years
by
recent
I mean
since
2020
there's
some
interesting
works
in that
regard
trying
to
formulate
the
interdependencies

and
emergence
with
mathematical
formalism
so
Lionel
Barnett
if I
remember
his
name
correctly
and
also
Anil
Seth
and
others
I forgot
some of
their
names
has done
some
pretty
interesting
work
in that
regard
in
recent
years
and
one
of
the
interesting

things
they've
came
across
is
that
they've
tried
to
apply
this
abstract
conception
of
interaction
and
interdependencies
from
information
theoretical
point
of
view
to
neuroimaging
data
and
so
they
they've
been able
to
describe
this
interaction
of
neural
interaction

between
different
brain
regions
without
accounting
explicitly
accounting
for
synaptic
or
I don't
know
junction
interaction
or
so on
so
it's
much
more
abstract
and
much
more
generalized
form
of
interaction
awesome
yes
anatomy

and
physiology
are
different
anatomy

is
describing
the
articulations
of the
system
but then
the
physiological
relationships
and correlations
and even
causes
are not
necessarily
following
along the
anatomical
lines
and
if one
wants the
anatomical
answer
you can
just
slice
the
tissue
and take
the
picture
so
something
different
is
happening
with

this
modeling
and
it's
important
like you
mentioned
that this
interaction
concept
amongst
variables
by the
time we're
labeling
variables
we're in
the map
we're not
labeling
the tissue
anymore
so the
only
possible
kind of
interaction
that these
variables
can have
is like
statistical
or
information
theoretic
but then
there's this
tantalizing

dream
slash
fallacy
slash
actionable
framework
all of
those
that
parameters
do map
on to
aspects of
the
territory
and that's
the map
territory
dialectic
some
maps
useful
in some
contexts
doesn't
reify
the
architecture
of the
map
and it
doesn't
prove or
disprove
any
architecture
of the
system

and the
interesting
thing is
that's not
even active
inference
specific
these are
active
inference
type
predictive
processing
predictive
processing
models
however
any
statistical
or
cognitive
modeling
approach
is going
to confront
this issue
map
and then
how do
we move
beyond
just
simply
saying
well
map
territory
means

we
can't
drive
or we
can't
do
route
planning
it's
like
well
if
dopamine
is
associated
with
precision
in this
model
where should
the deep
brain
stimulator
go
or what
kind of
food
would make
sense
to eat
or not
eat
and so
on
so we
can actually
reduce
uncertainty

about what
actions to
take
using maps
and that's
what these
are
and those
maps
arguably
works best
when
it's understood
what's a map
and what's the
territory
and understanding
what the map
is can be
fully
described
and held
to the
standards
of
reproducibility
whereas
holding
a territory
to those
same
epistemic
norms
or criteria
it
it doesn't
even make
sense

the territory
can't be
fully
described
but
when
they're
overlaid
even
symbolically
it can
become
ambiguous
ambiguous
well
next week
we're going
to
not record
so if
you're
going to be
there again
that's
awesome
otherwise
please fill
out the
feedback form
in the future
textbook group
section
and check
out the
project
ideas
but in
these

last
10
minutes
today
is there
any
sections
or
questions
on
chapter
five
that
anybody
wants
to look
at
or
would
people
like to
see
into
chapter
six
you
I've
just
added
those
additional
comments
I was
making
into
the
coda
and

I'm
making
a quick
visualization
of what
I was
thinking
about
I'm
not sure
if this
relates
directly
to the
model
but
just
I'll
need
a minute
to
get
this
done
thank
you
I'll
leave
the tab
up
we'll
return
to
it
any
preference
for a
section

of
five
or
to
look
ahead
to
chapter
six
in
the
recipe
even
though
we'll
return
to
that
in
like
a
month
well
looking
at
yes
go
ahead
Michael
I was
going to
say no
objection
here
I just
wanted
somebody
to

respond
okay
let's
revisit
the whole
first half
of the
book
which you
all have
now
completed
one or
more
passes
through
chapter
one
provided
an
overview
chapters
two
and
three
describe
the
low
road
and
the
high
road
the
low
road
how are
we

going to
do it
with
Bayesian
statistics
uncontentious
why
at the
very least
we need
repeated
measurement
if something
can't be
repeatedly
measured
by us
it's not a
thing to
us
and then a
little bit
more from
the inside
if something
can't take
adaptive
action
or more
adaptive
action to
persist
and repeatedly
measure
itself
it won't
be that
kind of

thing
either
so whether
you view
from the
outside
repeated
measurement
or you
view from
the inside
adaptive
action
we can
model both
of those
in terms
of this
self-evidencing
primary
imperative
how are we
going to
model that
self-evidencing
primary
imperative
Bayesian
statistics
chapter
four
with all
of its
shakes
and bumps
was describing
the generative
models of

active
inference
and how
variational
Bayesian
inference
is used
as a
heuristic
or an
approximation
to otherwise
intractable
Bayesian
statistical
models
and we
just
looked at
chapter
five
which
characterized
a few
subsystems
of the
mammalian
neuroanatomy
and then
linked
them
together
and
provided
some
references
to
other

models
that
go
into
a lot
more
detail
computationally
and
molecularly
here's
what
is going
to be
coming up
in the
second
half
of the
book
that
is
the
application
oriented
part
of the
book
now
it
may
not
go
all
the
way
to
the

guidebook
playbook
handbook
that
we
all
expect
and
prefer
but
it
is
more
pragmatically
and
application
oriented
than
the
first
half
of
the
book
in
chapter
six
there's
a
recipe
for
designing
active
inference
models
with
some
here

we've
even
gone
through
that
recipe
but
broadly
they're
going
to
lay
out
some
steps
about
systems
of
interest
about
specification
of
the
generative
model
and
the
generative
process
and
there's
a lot
of
good
discussions
around
even
broader

considerations

like

why

are

we

making

this

model

who

are

we

making

the

model

for

is

this

model

being

used

descriptively

like

historically

or is

it being

used

proactively

that's

the

recipe

so

you

want

to

open

up

a

restaurant

here's
the
active
inference
recipe
chapter
seven
and
chapter
eight
are
like
a
pair
chapter
seven
is
going
to
cover
generative
models
that
have
discrete
time
characteristics
chapter
eight
is
going
to
look
at
active
inference
in
continuous

time
discrete
time
active
inference
on the
top
of
figure
4.3
continuous
time
active
inference
generative
model
on the
bottom
of
figure
4.3
so
chapters
seven
and eight
are going
to
basically
zoom
in
on
figure
4.3
in
sequence
chapter
nine
model

based
data
analysis
is
going
to
describe
some
of
the
framings
and
challenges
and
opportunities
associated
with
integrating
empirical
data
with
these
kinds
of
generative
models
so
far
up
till
then
it's
kind
of
been
like
here's
the

generative
model
we
constructed
and
now
we're
going
to
simulate
its
behavior
so
generative
model
specification
simulation
in
silico
generation
of
synthetic
data
but
we're
not
always
just
interested
in
generating
synthetic
data
sometimes
we
want
to
take

in
empirical
data
from
measurements
and
that's
what
this
chapter
is
about
and
then
chapter
10
is
a
longer
chapter
there's
no
equations
or
figures
and
it
is
a
capstone
that's
describing
a lot
of the
conceptual
and
historical
context

and
as
kind
of
referenced
earlier
may
make
sense
to
even
read
first
that
is
where
we're
going
in
the
second
half
of
the
cohort
so
the
first
half
as
the
authors
lay out
is
epistemic
we're
getting
like

the
roads
into
the
city
what
the
city
is
about
or
one
restaurant
in
the
city
and
then
a
few
dishes
that
have
been
prepared
in
this
city
then
we're
going
to
go
behind
the
scenes
look
at

the
recipe
for
designing
those
models
talk
about
two
subtypes
of
cuisine
the
discrete
time
and
the
continuous
time
talk
about
connecting
these
generative
models
to
measurements
so
that
people
can
do
analyses
and
write
papers
and
all

of
that
make
things
that
do
things
in
the
world
and
then
having
done
all
of
that
chapter
10
connects
a lot
of
dots
that's
the
par
et
all
textbook
and
for
being
here
you
all
are
at
least

half
done
with
it
and
as
also
mentioned
we
go at
a
similar
pace
we
haven't
set
the
dates
yet
for
the
second
half
of
cohort
three
it'll
likely
pick up
again
in
May
when we
have
cohort
four
begin
with

chapter
one
and
if
you
have
any
feedback
or
ideas
it
would
be
extremely
appreciated
to
fill out
this
feedback
form
just
directly
in
CODA
you
can
just
submit
it
right
there
as
well
as
to
share
the
link

for
others
to
join
and
convey
to
them
just
say
hey
stick
around
you
will
learn
it's
not
XYZ
or
it
is
XYZ
but
feel
free
to
share
that
so
that
we
can
have
really
engaged
and
awesome

cohorts
such as
this
one
and
also
any
other
comments
you
want
to
write
there
or
any
thoughts
on
how
we
can
improve
it
and
next
week
we're
going
to
focus
all
about
that
we're
going
to
not
record

that
session
we're
going
to
talk
about
the
project
ideas
table
overall
reflections
on
the
first
half
next
steps
and
about
the
future
of
the
textbook
group

Yana
I'll
look
forward
to
your
image
and
in
the

last
minute
or
two
anyone
else
please
feel
free
to
add
other
comments
!
It's
not
exactly
goodbye
as we
will
meet
again
but
this
is
the
last
recording
for
this
half
of
the
cohort
!
so
this
is

not
the
only
web
we
can
draw
but
it's
a
really
important
one
for
certain
kinds
of
statistical
models
next
week
same
time
for
feedback
and
next
steps
thank
you
everybody
see
you
then